UNDERWATER DRONE FOR SEARCH IN SHALLOW WATERS

PROJECT REPORT

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of

Bachelor of Technology

In

Mechatronics

Department of Mechatronics

Nirmala College of Engineering, Technology & Management

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DECLARATION

We undersigned hereby declare that the project report on "**UNDERWATER DRONE FOR SEARCH IN SHALLOW WATERS**", submitted for partial fulfilment of the requirements for the award of degree of Bachelor of Technology of the APJ Abdul Kalam Technological University,Kerala is a bonafide work done by us under supervision of **Mr. JAISON C V, Assistant Professor, Department of Mechatronics**. This submission represents our ideas in our own wordsand where ideas or words of others have been included, we have adequately and accurately cited and referenced the original sources. We also declare that we have adhered to ethics of academic honesty and integrity and have not misrepresented or fabricated any data or idea or fact or source in my submission. We understand that any violation of the above will be a causefor disciplinary action by the institute and/or the University and can also evoke penal action from the sources which have thus not been properly cited or from whom proper permission hasnot been obtained.

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CERTIFICATE

This is to certify that the work which is being presented in the B.Tech Seminar Report entitled **"UNDERWATER DRONE FOR SEARCH IN SHALLOW WATERS"** submitted in partial fulfilment of the requirements for the award of the **Bachelor of Technology in Mechatronics** and submitted by **Akin Ponnachen, Alvit K V, Anand S, Joyal Wilson,** to the Department of Mechatronics of Nirmala College of Engineering, Technology & Management, Chalakudy, Thrissur, is an authentic record of my own work under the supervision of **Mr. Jaison C V, Asst. Professor, Department Of Mechatronics, Nirmala College of Engineering, Technology & Management**. The matter presented in this has not been submitted for the award of any other degree elsewhere.This is to certify that the above statement made by the candidates are correct to the best of my knowledge.

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ABSTRACT

Underwater drones, also known as remotely operated underwater vehicles (ROVs). These devices comprise three main parts: a surface control unit, a tether, and the drone itself. The surface control unit facilitates communication and control, while the tether transmits power and data between the drone and the surface. The drone is equipped with cameras, sensors, and thrusters for navigation and data collection. The working of underwater drones involves operators remotely piloting them from the surface, allowing access to underwater locations. Their high ability to navigate and to withstand pressure make them indispensable for underwater research, exploration, search and rescue operation. Advantages of underwater drones include reduced risk to human divers, and the collection of crucial data for scientific research and environmental monitoring, ultimately contributing to our understanding of the world's oceans.

KEYWORDS: Remotely operated underwater vehicles (ROVs), Exploration, Cameras.

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ABBREVIATIONS

ROV	-	Remotely Operated Vehicle
DC	-	Remotely Operated Vehicle
MPRL	-	Model Predictive Resource Leveling
GPIO	-	General Purpose Input/Output
UAV	-	Unmanned Aerial Vehicles
LAN	-	Local Area Network
CSMA/CD Detection	-	Carrier Sense Multiple Access with Collision
MAC	-	Media Access Control
BLDC	-	Brush-Less Direct Current
ESC	-	Electronic Speed Control
BMS	-	Battery Management System
NTC	-	Negative Temperature Coefficient
GPS	-	Global Positioning System

CHAPTER 1

INTRODUCTION

The design and fabrication of Remotely Operated Vehicles (ROVs) represent a cutting edge field at the intersection of engineering and marine exploration. ROVs are unmanned, submersible vehicles that play a pivotal role in conducting underwater operations in environments too hazardous or inaccessible for human divers. This field encompasses a diverse range of applications, from underwater research and environmental monitoring to offshore oil and gas exploration and maintenance of subsea infrastructure. The design phase involves the careful consideration of structural integrity, buoyancy, and propulsion systems, while the fabrication process demands precision engineering to ensure the ROV can withstand the harsh conditions of the underwater world. Innovations in materials, control systems, and imaging technology contribute to the continual evolution of ROV design, making it an exciting and dynamic area of research and development.

1.1 OBJECTIVE

- To achieve real-time monitoring under shallow waters with minimum turbulence.
- To produce a low-budget underwater drone for local use.
- To find lost items as well as help recover drowned bodies for ease of divers.

There are many cases of people drowned to death but the body was stuck underwater. There was an incident of drowning of a teenager whose body was stuck underwater near a rock, and the recovery was paused due to lack of light. Using these drones can overcome this limitation, as it consists of light as well as sensors to detect the body or objects.

CHAPTER 2

LITERATURE REVIEW

[1] ***Design and testing of a compact inexpensive prototype remotely operated underwater vehicle for shallow water operation" by a. Tanveer and S. M. Ahmad***

The article introduces the design and real-time testing of an inspection class ROV, that weighs1.59 kg. The envisioned ROV can move in three axes of translation, ascent/descent, and rotation. The ROV was designed to reach depths of up to 100 meters, addressing the problem of human divers who can only reach depths of 30 meters. Moreover, the ROV is propelled by three DC motors controlled by an MPRL PI controller. Improved ROV stability is achieved by using a complementary filter to reduce noise from the MPU 9250 sensor. The ROV is programmed using open-source software tools, allowing for the incorporation of additional sensors and functionality as per the operational requirements. Additionally, the adaptability of mechanical design and inexpensive hardware expands the range of possible applications, including monitoring and the research of maritime flora and fauna, without sacrificing the accuracy of the data collected.

[2] ***"Life cycle management of the 3d printer technology to design an underwater drone hull to study the arctic" by E V Kozlova1 , Yakinchuk ,K A Starikov***

The analysis of the current state of the additive technology market revealed the following: additive production remains a relatively new application field. The USA, China, Germany, France and certain entities of the United Kingdom remain leading positions in the sphere of additive technologies. However, the Russian Federation also has the technological capacity to create 3D printers of various configurations. Production of some of such printers has already been launched and operates successfully. All those devices that use additive manufacturing technologies have practically no limitations on the geometry of the manufactured product, and the speed of manufacture of components does not depend on their complexity, which results in significant economic benefits. These technologies revolutionize product design, production and procurement. They affect the entire product life cycle, which is a strategic area of interest for business executives.The study revealed the fact that graph construction provides an opportunity to describe a complex sociotechnical system (a 3D printer technology) and assess the most efficient ways to design its life cycle (resources and time spent were taken into consideration). In terms of modern technology development and automation of production

processes, PLM and PDM solutions provide for the efficient life cycle management of a product. A complete switch to additive technologies is not yet possible. Therefore, the problem of production and operation of 3D printers of different configurations remains a topical one and requires further consideration.

[3] ***"Investigating the effects of coating systems on the degradation behavior of 3d printed pressure vessels" by Birendra Chaudhary, Hewenxuan Li***

This study investigated the impact of seawater infusion and weathering on 3D printed PLA composites, both with and without a coating system. Similarly, implosion experiments were conducted on both aged specimens with and without coating systems using a high pressure tank facility. After completing the work, the following conclusions and recommendations can be made: The application of coatings offered a protective layer that effectively mitigated mass absorption and minimized stiffness degradation in the structures. The higher activation energy observed in coated PLA composites indicated a slower diffusion and degradation process than the base PLA composites. Among the evaluated coatings, the Epoxy + Silicone coating exhibited the lowest water absorption and was identified as a suitable option for sea structures.3D printed structures have fast diffusivity rates, emphasizing the need for caution when utilizing such structures in long term underwater applications.

[4] ***"A review of recent advancements in sensors employed in unmanned underwater vehicles" by Emily Shields and Masudul H Imtiaz***

Many underwater vehicles use acoustic and magnetic sensors for navigation and research purposes. The acoustic sensors used for navigation are baseline sensors. Baseline sensing technology uses a network of sea floor mounted baseline transponders to communicate an underwater drone's accurate position fix using those transponders as reference points. The use of a variety of sonar systems can assist these underwater vehicles in scanning the ocean floor, detecting the topography of the sea floor, or recognizing objects at extreme depths. Magnetic sensors can do a similar function with object recognition. Large cables are run underwater on the ocean floor, and magnetic sensors can help guide the ROVs and AUVs to track and inspect this cable. Magnetic sensors used for research can track these cables and collect data in the presence of sensor noise and ocean currents [16]. Navigationally, magnetic sensors can detect changes in the Earth's magnetic field to determine location. Understanding how underwater drone sensing technology has developed over the years is important. Acoustic and magnetic

sensors have been used for navigation and data collection for many years. Thus, their evolution is integral to predicting the further trajectory of acoustic and magnetic sensing technology.

[5] *"Design and construction of an rov for underwater exploration" by Oscar Adrian Aguirre Castro, Everardo Inzunza González*

The ROV development and implementation contained in a box of size 18.41 cm × 29.50 cm ×33.50 cm, with a weight of 15.64 kg, is introduced. The proposed ROV performs the translational, ascent, descent, Sensors 2019, 19, 5387 22 of 25 and rotational movements on three axes to capture images of 800 × 640 pixels on video graphic array standard. The ROV design was done to reach up to 100 m underwater, which can solve the problem of divers who can only reach 30 m. The motion control, 3D position, temperature sensing, and video capture are performed in parallel by using the threading library, and they are processed by a main algorithm that was programmed to use the four cores of a SoC Raspberry Pi 3. The communication between the ROV and the remote control is handled by a graphical user interface coded on python, which is suitable for different operating systems, such as GNU/Linux, Windows, Android, and OS X. Furthermore, the ROV moves under the six brushless motors governed through a smart PID controller. From experimental results,the brushless motors were calibrated to work with a short pulse width of 1 µs to improve the ROV.

CHAPTER 3

DESIGN OF UNDERWATER DRONE

This design utilizes five BLDC motors for multi-directional movement underwater. Three vertical motors control upward and downward motion, providing stability and depth adjustment. Two horizontal motors enable forward, backward, and sideways movement, ensuring dynamic maneuverability. This configuration optimizes control and agility, allowing the drone to navigate efficiently in diverse underwater environments. The vertical alignment enhances stability during ascent and descent, while the horizontal arrangement facilitates precise lateral and longitudinal movement. Overall, the combination of vertical and horizontal BLDC motors provides versatility, allowing the underwater drone to perform a wide range of tasks effectively, from exploration to surveillance and research missions in aquatic environments.

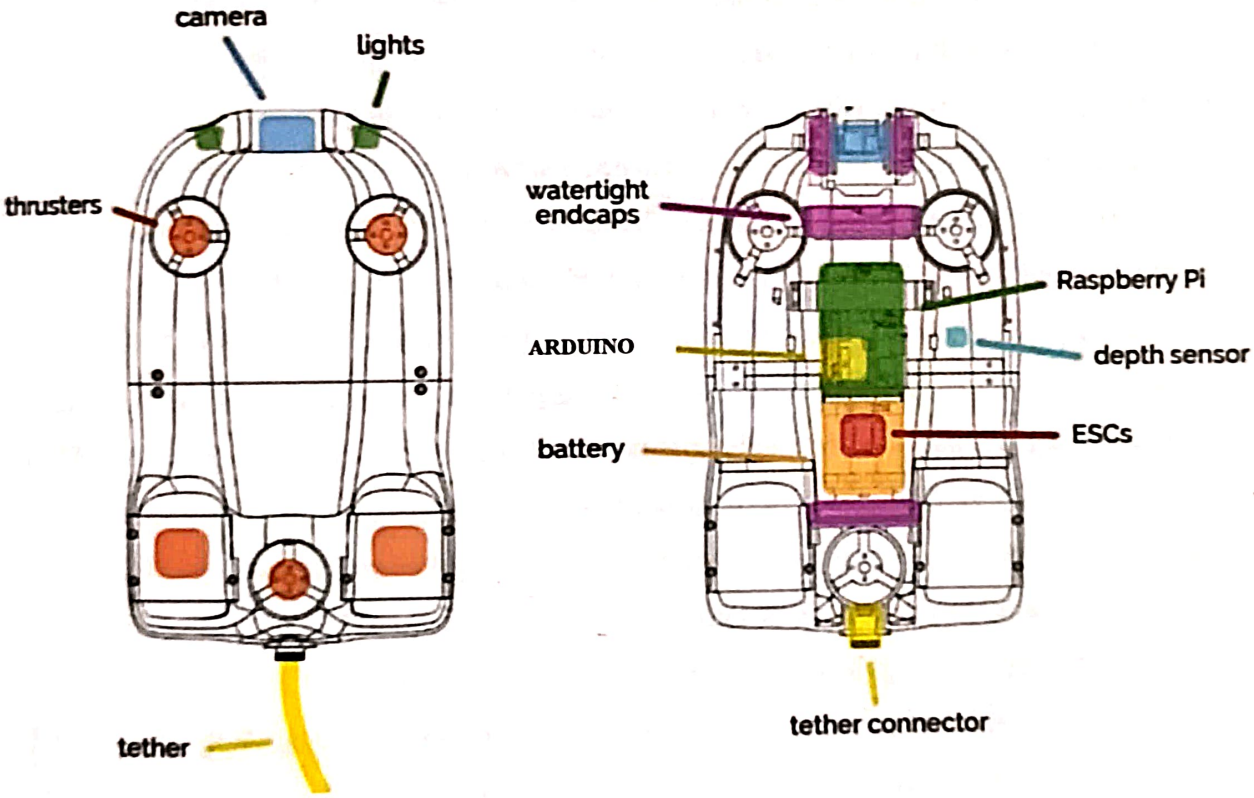

Fig 3.1 Design of Underwater Drone

CHAPTER 4

CIRCUIT DIAGRAM OF UNDERWATER DRONE

4.1 Arduino :

It runs the ArduSub firmware. The Arduino is connected to the 5 BLDC motors & controlled with an ESC motor. The Arduino Uno is an open-source microcontroller board based on the Microchip ATmega328P microcontroller (MCU) and developed by Arduino.cc and initially released in 2010. The board is equipped with sets of digital and analog input/output (I/O) pins that may be interfaced to various expansion boards (shields) and other circuits. The board has 14 digital I/O pins (six capable of PWM output), 6 analog I/O pins, and is programmable with the Arduino IDE USB cable. It can be powered by a USB cable or a barrel connector that accepts voltages between 7 and 20 volts, such as a rectangular 9-volt battery. The directional data is given by the Raspberry Pi computer to the Arduino board.

4.2 Raspberry Pi:

The Companion Computer streams video to the Topside Computer, and relays MAVLink communications between the autopilot and the Topside Computer via Ethernet. The Raspberry Pi 4 is a credit card-sized single-board computer developed by the Raspberry Pi Foundation. It is part of the Raspberry Pi series, which consists of affordable, programmable computers intended for educational and DIY (do-it-yourself) purposes. The Raspberry Pi 4 Model B is a popular single-board computer developed by the Raspberry Pi Foundation. It features a quad-core ARM Cortex-A72 processor, ranging from 1GB to 8GB of RAM, USB 3.0 ports, dual-band 802.11ac Wi-Fi, Bluetooth 5.0, Gigabit Ethernet, dual micro HDMI ports supporting up to 4K resolution, and a microSD card slot for storage. Powered by a quad-core ARM Cortex-A72 processor running at 1.5GHz, it offers options for 1GB, 2GB, 4GB, or 8GB of LPDDR4-3200 SDRAM, ensuring smooth performance for various applications. It features dual-band 802.11ac wireless LAN, Bluetooth 5.0, Gigabit Ethernet, and a plethora of ports including two USB 3.0 and two USB 2.0 ports, two micro HDMI ports supporting up to 4Kp60 resolution, a 3.5mm audio jack, and a microSD card slot for storage. Its 40-pin GPIO header remains compatible with previous Raspberry Pi boards, enabling seamless integration with a wide array of peripherals. This, coupled with its support for various Linux distributions and third-party operating systems, makes the Raspberry Pi 4 Model B a go-to choice for projects ranging from basic tinkering to complex robotics and IoT applications.It's widely used for various projects ranging from simple electronics to home automation and robotics due to its versatility and affordability.The Raspberry Pi 4, released in

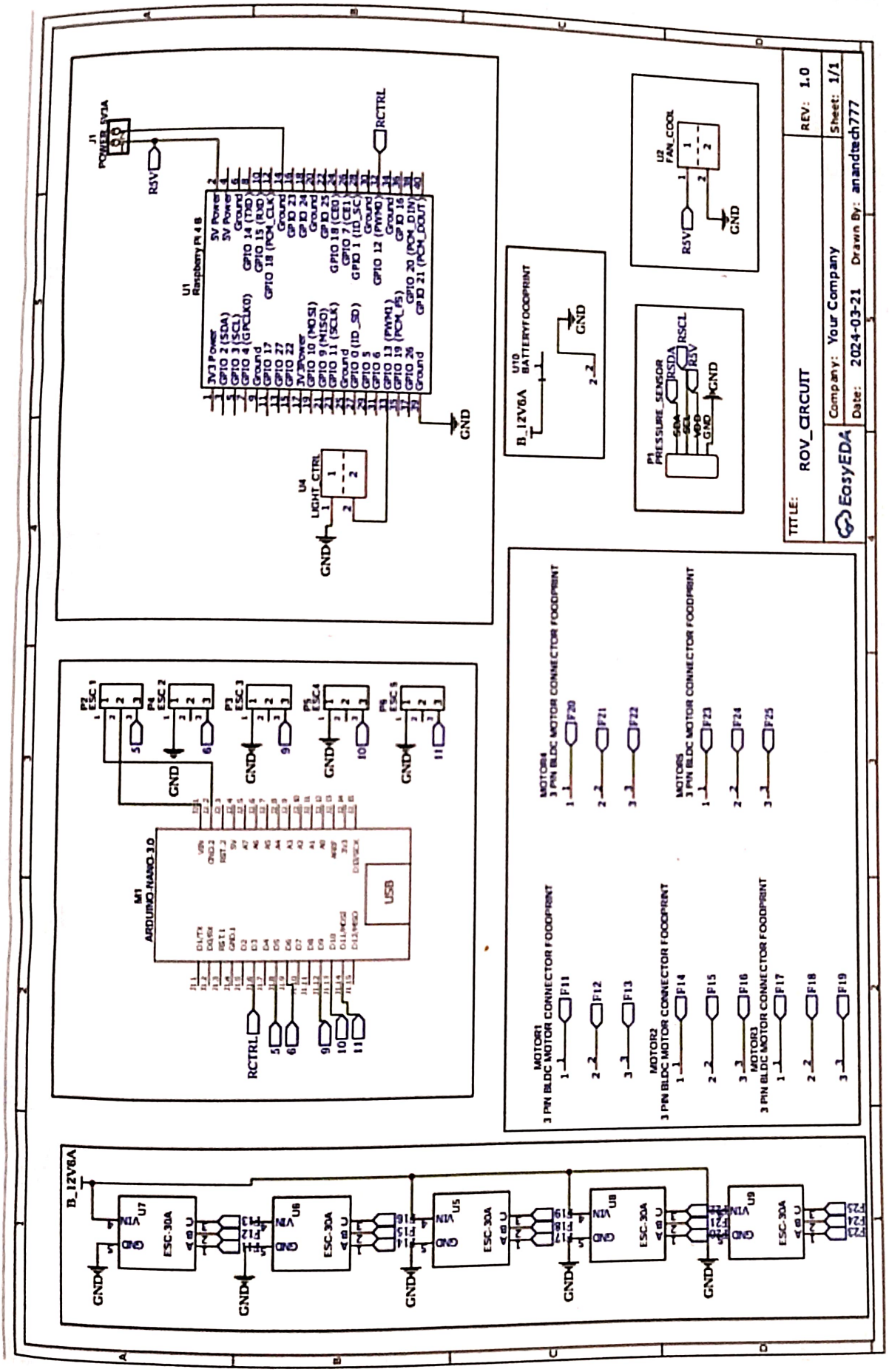

Fig: 4.1 Circuit diagram.

2019, features a quad-core ARM Cortex-A72 processor, options for 1GB, 2GB, or 4GB of RAM, multiple USB ports, micro-HDMI ports for video output, a Gigabit Ethernet port, and GPIO (General Purpose Input/Output) pins for hardware interfacing. The Raspberry Pi is connected to the Arduino for directional movement and receive signals from user interface.

4.3 Camera:

The camera allows the pilot to see and record from the vehicle's point of view. If paired with a Camera Tilt Mount or other gimbal it can be moved during operation. One of the most remarkable features of this cutting-edge camera is its remarkable focus range, spanning from 4 to 20 meters. This expansive range empowers users to capture images with unparalleled clarity and depth, whether close or at a distance. From macro photography to long-range surveillance, the 25-megapixel USB camera delivers exceptional performance across a spectrum of scenarios. The camera is connected to the Raspberry pi for real time monitoring.

4.4 Electronic Speed Controls (ESCs):

ESCs are used to control the speed/thrust of motors and thrusters. The term "ESC 30A" refers to an Electronic Speed Controller with a current rating of 30 amperes. In the context of electric motor systems, particularly in the realm of radio-controlled vehicles and drones, an ESC plays a crucial role in regulating the speed and direction of an electric motor. The "30A" designation indicates the maximum current capacity that this specific ESC can handle, which is 30 amperes. This parameter is essential as it defines the maximum amount of electrical current that can flow through the ESC without causing damage. ESCs are commonly used in various applications, such as hobbyist drones, remote-controlled cars, boats, and other electric-powered vehicles. They receive signals from a control unit, such as a radio transmitter, and translate these signals into adjustments in the motor's speed, allowing for precise control and maneuverability in the respective application. ESCs are connected to Arduino and BLDC motors for movement.

4.5 Thrusters:

Thrusters are necessary to maneuver an underwater vehicle around. The number and orientation of thrusters on a vehicle determines the number of degrees of freedom (DoF) it may maneuver in. The A2212 1400Kv brush-less DC (BLDC) motor is a specific type of electric motor designed for various applications, particularly in the realm of remote-controlled aircraft, drones, and other hobbyist projects. The term "A2212" refers to the motor's model or series, while "1400Kv" denotes the motor's speed constant, indicating that it generates 1400 RPM per volt under no load conditions. This BLDC motor operates without brushes, which reduces

friction and wear, leading to improved durability and efficiency. The A2212 1400Kv motor typically features a compact design, high power-to-weight ratio, and compatibility with electronic speed controllers (ESC) for precise speed and direction control. The motors are connected to ESCs.

4.6 Power Supply: The power supply and distribution system powers all the onboard electronics, including the high current draw of the thrusters. A regulator converts the main supplied power (from a battery or otherwise) into appropriate voltages for the more sensitive electronics it must ensure a steady supply to avoid the computers restarting and losing control of the vehicle. The 12v Battery are connected 5v voltage converter, and which is connected to the Arduino and raspberry pi. A Li-ion 18650 battery is a rechargeable lithium-ion cylindrical cell that measures 18 millimeters in diameter and 65 millimeters in length. Widely used in various electronic devices, these batteries have become a popular choice due to their high energy density, lightweight nature, and ability to deliver a consistent power output. The "18650" designation refers to the dimensions of the battery, making it a standardized form factor. Li-ion 18650 batteries utilize lithium ions as the charge carriers, shuttling between the positive and negative electrodes during charging and discharging cycles. Their rechargeable nature makes them cost-effective and environmentally friendly compared to single-use batteries. The BMS (Battery Management System) are used manage the discharge and charging of the battery. The 12v Battery are connected to the motor driver (ESC).

4.7 Tether: A tether is a length of cable which connects the Companion Computer to the Topside Computer. Frames, or packets of data, are encapsulated with source and destination MAC addresses. Ethernet commonly utilizes twisted pair cables with RJ45 connectors, and its speed has evolved from the original 10 Mbps (Ethernet) to gigabit (1000 Mbps) and beyond. Modern Ethernet standards include Fast Ethernet (100 Mbps), Gigabit Ethernet, and 10 Gigabit Ethernet, providing increased data transfer rates. Ethernet is foundational for local network communication, supporting internet connectivity, file sharing, and various applications. It allows low latency high bandwidth communication to a vehicle in water, which other technologies are poorly suited to. The ethernet cable which is connected to the raspberry pi and user computer.

CHAPTER 5

WORKING PRINCIPLE OF UNDERWATER DRONE

5.1 Autopilot(ARDUINO UNO) :

The Brain of the ROV. The Autopilot(Arduino UNO), or flight controller board, serves as the central processing unit of the ROV. Running on ArduSub firmware, it processes pilot inputs and sensor data to control motors, lights, servos, and relays. Its precise control ensures stable navigation and operation in underwater environments. The Arduino Uno is an open-source microcontroller board based on the Microchip ATmega328P microcontroller (MCU) and developed by Arduino.cc and initially released in 2010. The board is equipped with sets of digital and analog input/output (I/O) pins that may be interfaced to various expansion boards (shields) and other circuits. The board has 14 digital I/O pins (six capable of PWM output), 6 analog I/O pins, and is programmable with the Arduino IDE (Integrated Development Environment), via a type B USB cable. It can be powered by a USB cable or a barrel connector that accepts voltages between 7 and 20 volts, such as a rectangular 9-volt battery

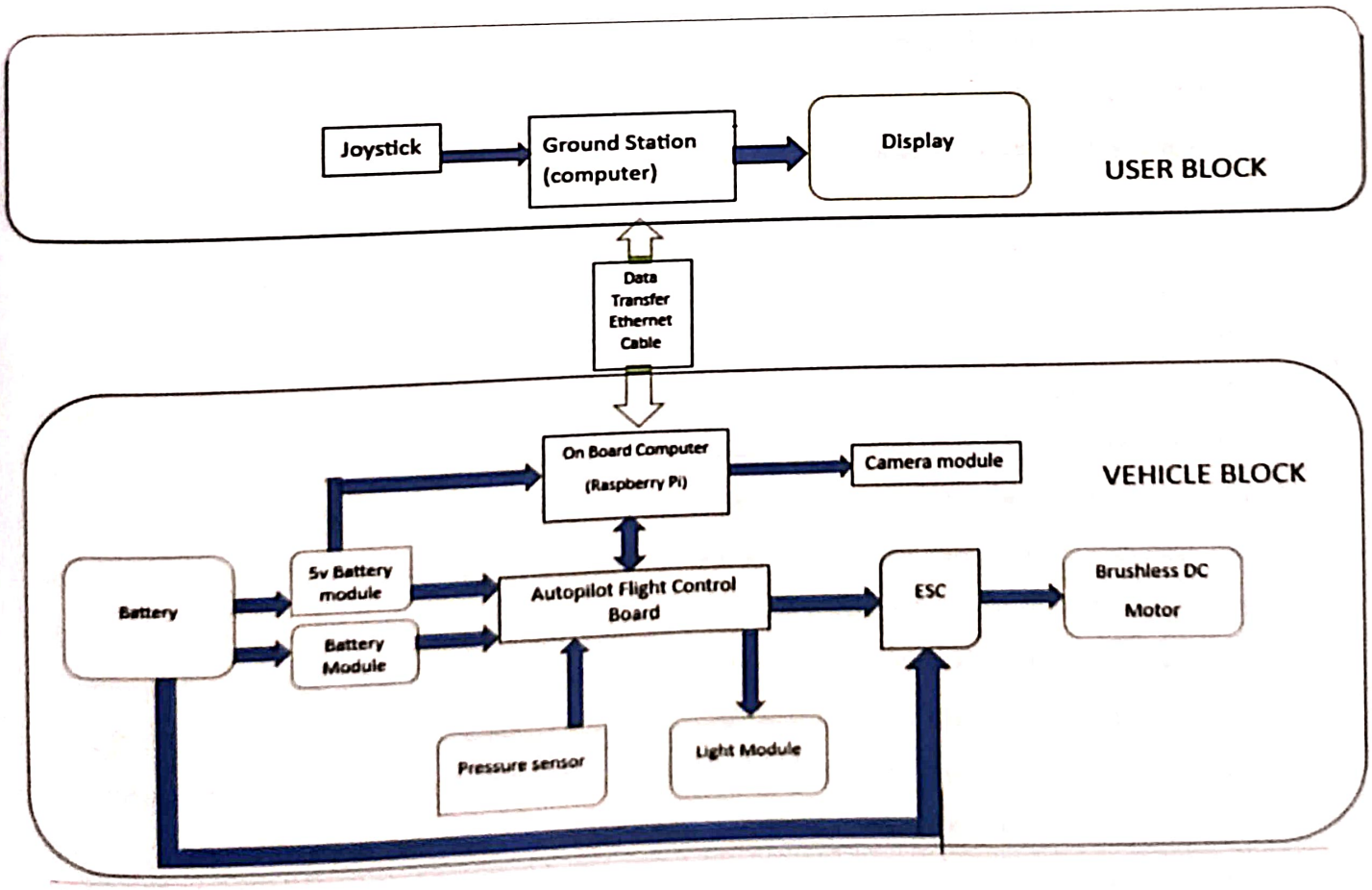

Fig: 5.1 Block diagram of ROV.

5.2 Companion Computer:

Enabling Communication. The Companion Computer facilitates communication between the ROV and the surface operator. The Raspberry Pi 4 is a credit card-sized single-board computer developed by the Raspberry Pi Foundation. It is part of the Raspberry Pi series, which consists of affordable, programmable computers intended for educational and DIY (do-it-yourself) purposes. The Raspberry Pi 4, released in 2019, features a quad-core ARM Cortex-A72 processor, options for 1GB, 2GB, or 4GB of RAM, multiple USB ports, micro-HDMI ports for video output, a Gigabit Ethernet port, and GPIO (General Purpose Input/Output) pins for hardware interfacing. It streams live video feeds and relays MAVLink communications, ensuring seamless control and monitoring of the vehicle's operations from the surface.

5.3 Camera System:

Visual Feedback for Operators. The Camera system provides visual feedback to operators, allowing them to see and record the underwater environment from the ROV's perspective. One of the most remarkable features of this cutting-edge camera is its remarkable focus range, spanning from 4 to 20 meters. This expansive range empowers users to capture images with unparalleled clarity and depth, whether close or at a distance. From macro photography to long-range surveillance, the 25-megapixel USB camera delivers exceptional performance across a spectrum of scenarios. Paired with tilt mounts or gimbals, it enhances situational awareness and aids in precise maneuvering.

5.4 Electronic Speed Controls (ESCs):

Regulating Thrust. ESCs regulate the speed and thrust of the ROV's motors and thrusters, enabling precise control of movement underwater. The term "ESC 30A" refers to an Electronic Speed Controller with a current rating of 30 amperes. In the context of electric motor systems, particularly in the realm of radio-controlled vehicles and drones, an ESC plays a crucial role in regulating the speed and direction of an electric motor. The "30A" designation indicates the maximum current capacity that this specific ESC can handle, which is 30 amperes. This parameter is essential as it defines the maximum amount of electrical current that can flow through the ESC without causing damage. ESCs are commonly used in various applications, such as hobbyist drones, remote-controlled cars, boats, and other electric-powered vehicles. They receive signals from a control unit, such as a radio transmitter, and translate these signals into adjustments in the motor's speed, allowing for precise control and maneuverability in the respective application. Operators can adjust ESCs to maneuver the ROV with accuracy, making it suitable for various tasks.

5.5 Thrusters:

Propulsion for Maneuverability. Thrusters are essential propulsion systems that enable the ROV to maneuver in three-dimensional space underwater. The A2212 1400Kv brush-less DC (BLDC) motor is a specific type of electric motor designed for various applications, particularly in the realm of remote-controlled aircraft, drones, and other hobbyist projects. The term "A2212" refers to the motor's model or series, while "1400Kv" denotes the motor's speed constant, indicating that it generates 1400 RPM per volt under no load conditions. This BLDC motor operates without brushes, which reduces friction and wear, leading to improved durability and efficiency. The A2212 1400Kv motor typically features a compact design, high power-to-weight ratio, and compatibility with electronic speed controllers (ESC) for precise speed and direction control. The number and orientation of thrusters determine the vehicle's degrees of freedom, allowing agile movement in all directions.

5.6 Power Sensing Module:

Monitoring Power Consumption. The Power Sensing Module provides critical information about power consumption and battery levels to the autopilot. A Li-ion 18650 battery is a rechargeable lithium-ion cylindrical cell that measures 18 millimeters in diameter and 65 millimeters in length. Widely used in various electronic devices, these batteries have become a popular choice due to their high energy density, lightweight nature, and ability to deliver a consistent power output. The "18650" designation refers to the dimensions of the battery, making it a standardized form factor. Li-ion 18650 batteries utilize lithium ions as the charge carriers, shuttling between the positive and negative electrodes during charging and discharging cycles. Their rechargeable nature makes them cost-effective and environmentally friendly compared to single-use batteries. Monitoring power usage optimizes performance and prevents unexpected shutdowns or malfunctions.

5.7 Power Supply and Distribution Systems:

Ensuring Reliability. Power Supply and Distribution Systems ensure reliable operation of all onboard electronics, including thrusters. A Battery Management System (BMS) is a crucial component in managing the performance, safety, and longevity of rechargeable batteries. The specifications of a BMS can vary depending on the application and the type of batteries it is designed to manage.A Battery Management System (BMS) is a crucial component designed to monitor, control, and safeguard rechargeable batteries. Its primary function is to ensure the efficient and safe operation of the battery pack. The BMS accomplishes this through a set of specifications and features.At its core, the BMS oversees the individual cells within the battery

pack, managing factors such as voltage, current, and temperature. Regulators convert the main power supply into appropriate voltages, ensuring steady power to sensitive electronics to prevent system failures.

5.8 Tether:

A Lifeline for Communication. The Tether connects the Companion Computer to the surface computer, enabling low-latency, high-bandwidth communication underwater. Frames, or packets of data, are encapsulated with source and destination MAC addresses. Ethernet commonly utilizes twisted pair cables with RJ45 connectors, and its speed has evolved from the original 10 Mbps (Ethernet) to gigabit (1000 Mbps) and beyond. Modern Ethernet standards include Fast Ethernet (100 Mbps), Gigabit Ethernet, and 10 Gigabit Ethernet, providing increased data transfer rates. Ethernet is foundational for local network communication, supporting internet connectivity, file sharing, and various applications. Unlike wireless technologies, tethers provide reliable communication over long distances, ensuring continuous control and monitoring of the ROV.

CHAPTER 6

COMPONENTS OF UNDERWATER DRONE

6.1 RASPBERRY PI 4

The Raspberry Pi 4 is a credit card-sized single-board computer developed by the Raspberry Pi Foundation. It is part of the Raspberry Pi series, which consists of affordable, programmable computers intended for educational and DIY (do-it-yourself) purposes. The Raspberry Pi 4, released in 2019, features a quad-core ARM Cortex-A72 processor, options for 1GB, 2GB, or 4GB of RAM, multiple USB ports, micro-HDMI ports for video output, a Gigabit Ethernet port, and GPIO (General Purpose Input/Output) pins for hardware interfacing. It has become popular for a variety of applications, including as a low-cost desktop computer, media center, home server, and a platform for learning programming and electronics.

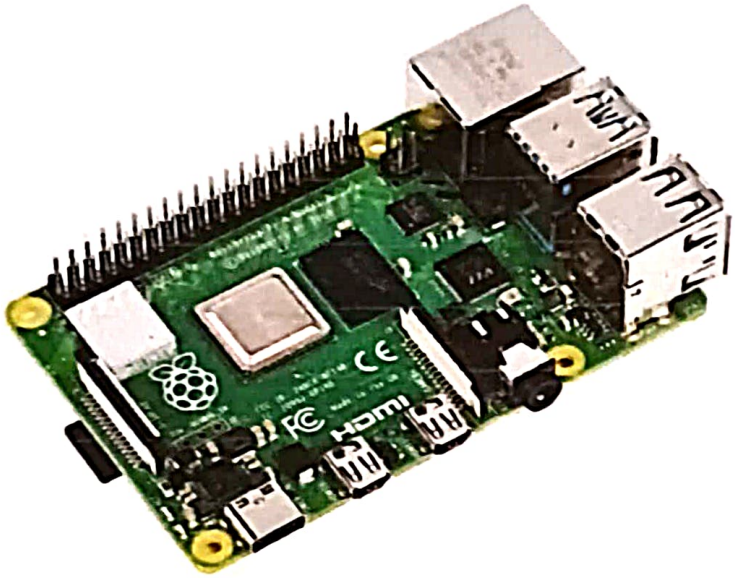

Fig:6.1 Raspberry pi 4

Specifications of Raspberry Pi 4

- Broadcom BCM2711, Quad-core Cortex-A72 64-bit SoC
- 4GB LPDDR4-3200 SDRAM
- Gigabit Ethernet
- 2 x USB 3.0 ports, 2 x USB 2.0 ports
- Operating power 5V 3A

6.2 ARDUINO UNO

The Arduino Uno is an open-source microcontroller board based on the Microchip

ATmega328P microcontroller (MCU) and developed by Arduino.cc and initially released in 2010. The board is equipped with sets of digital and analog input/output (I/O) pins that may be interfaced to various expansion boards (shields) and other circuits. The board has 14 digital I/O pins (six capable of PWM output), 6 analog I/O pins, and is programmable with the Arduino IDE (Integrated Development Environment), via a type B USB cable. It can be powered by a USB cable or a barrel connector that accepts voltages between 7 and 20 volts, such as a rectangular 9-volt battery.

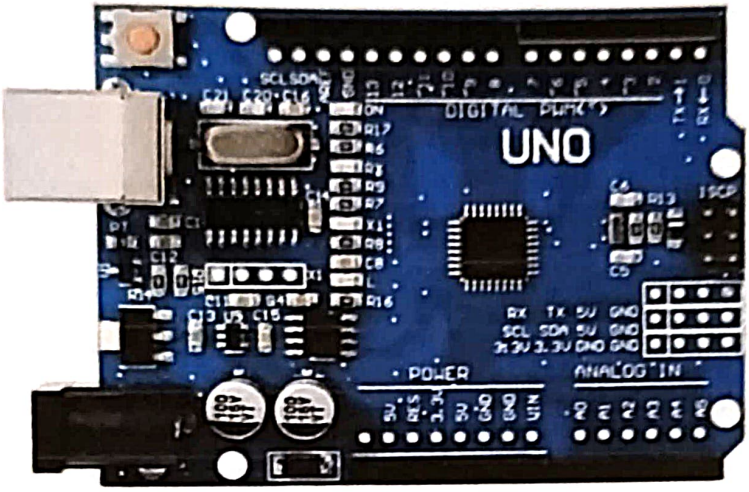

Fig:6.2 Arduino UNO

Specifications of Arduino UNO

- IC: Microchip ATmega328P
- Operating Voltage: 5 Volts
- DC Current per I/O Pin: 20 mA
- DC Current for 3.3V Pin: 50mA
- Size: 68.6 mm x 53.4 mm
- Weight: 25 g

6.3 ETHERNET COMMUNICATION

Ethernet communication is a widely used networking technology that enables devices to communicate within a local area network (LAN). It operates on the principles of wired connections, employing a protocol for data transmission called CSMA/CD (Carrier Sense Multiple Access with Collision Detection). In this method, devices listen to the network before transmitting data to avoid collisions. Ethernet uses a star or bus topology, connecting devices through switches or hubs. Each device on the network has a unique MAC (Media Access Control) address, facilitating data routing to the intended recipient.

Fig: 6.3 Ether-net cable

Frames, or packets of data, are encapsulated with source and destination MAC addresses. Ethernet commonly utilizes twisted pair cables with RJ45 connectors, and its speed has evolved from the original 10 Mbps (Ethernet) to gigabit (1000 Mbps) and beyond. Modern Ethernet standards include Fast Ethernet (100 Mbps), Gigabit Ethernet, and 10 Gigabit Ethernet, providing increased data transfer rates. Ethernet is foundational for local network communication, supporting internet connectivity, file sharing, and various applications. Its reliability, scalability, and widespread adoption make it a fundamental technology in today's interconnected world.

6.4 BLDC MOTOR A2212 1400Kv

The A2212 1400Kv brush-less DC (BLDC) motor is a specific type of electric motor designed for various applications, particularly in the realm of remote-controlled aircraft, drones, and other hobbyist projects. The term "A2212" refers to the motor's model or series, while "1400Kv" denotes the motor's speed constant, indicating that it generates 1400 RPM per volt under no load conditions. This BLDC motor operates without brushes, which reduces friction and wear, leading to improved durability and efficiency. The A2212 1400Kv motor typically features a compact design, high power-to-weight ratio, and compatibility with electronic speed controllers (ESC) for precise speed and direction control. Its popularity in hobbyist communities stems from its ability to deliver reliable and responsive performance, making it well-suited for applications where compact, lightweight.

Fig: 6.4 BLDC Motor.

Specifications of BLDC motor

- Number of cells: 2 3s Lipo; 6 10 cell NiMh
- Maxium efficiency current: 6 12A (>75%) No load current (10v): 0.7A
- Maximum current: 16A/60s
- Maximum efficiency: 78%
- Internal resistance: 65mΩ
- Poles: 14
- Maximum watts: 180W
- Motor bolt pattern (back of motor): 16mm x 19mm (hole to hole) X mount hole pattern: 32mm x 32mm (hole to hole)

Dimensions of BLDC motor

- Motor size: 27.5mm x 30mm
- Shaft size: 3.17mm
- Weight: 47

6.5 ESC 30A

The term "ESC 30A" refers to an Electronic Speed Controller with a current rating of 30 amperes. In the context of electric motor systems, particularly in the realm of radio-controlled vehicles and drones, an ESC plays a crucial role in regulating the speed and direction of an electric motor. The "30A" designation indicates the maximum current capacity that this specific ESC can handle, which is 30 amperes. This parameter is essential as it defines the maximum amount of electrical current that can flow through the ESC without causing damage. ESCs are commonly used in various applications, such as hobbyist drones, remote-controlled cars, boats, and other electric-powered vehicles. They receive signals from a control unit, such as a radio transmitter, and translate these signals into adjustments in the motor's speed, allowing for precise control and maneuverability in the respective application. The ESC 30A, with its 30-ampere capacity, is suitable for powering motors with moderate power requirements, making it a versatile component in the world of remote-controlled and electrically driven vehicles.

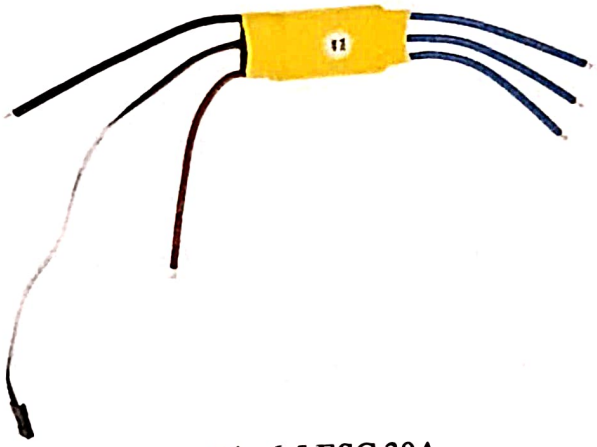

Fig:6.5 ESC 30A.

Specifications of ESC 30A

- Model SIMONK 30A
- Current: 30A
- Voltage:12V
- Length: 34 mm
- Width: 24 mm
- Height: 9 mm
- Weight: 23g

6.6 PROPELLER DESIGN

Fig:6.6 Propeller

Propeller Design:

- Diameter: 70mm
- Pitch: 60mm
- Blade Number: 3
- Twist: 10 degrees from root to tip
- 3D Printing: Consider layer resolution, orientation, and use a strong and durable material suitable for underwater use.
- Balancing: Ensure the propeller is well balanced to avoid vibrations.

6.7 BATTERY

A Li-ion 18650 battery is a rechargeable lithium-ion cylindrical cell that measures 18 millimeters in diameter and 65 millimeters in length. Widely used in various electronic devices, these batteries have become a popular choice due to their high energy density, lightweight nature, and ability to deliver a consistent power output. The "18650" designation refers to the dimensions of the battery, making it a standardized form factor. Li-ion 18650 batteries utilize lithium ions as the charge carriers, shuttling between the positive and negative electrodes during charging and discharging cycles. Their rechargeable nature makes them cost-effective and environmentally friendly compared to single-use batteries. These batteries find applications in a range of devices, from portable electronics like laptops and cameras to electric vehicles and energy storage systems, contributing to the advancement of portable technology

and sustainable energy solutions.

Fig:6.7 Li-ion 18650 battery.

Specification of Li-ion 18650 battery.

- Voltage 3.7 V
- Current 2A

6.8 BATTERY MANAGEMENT SYSTEM (BMS)

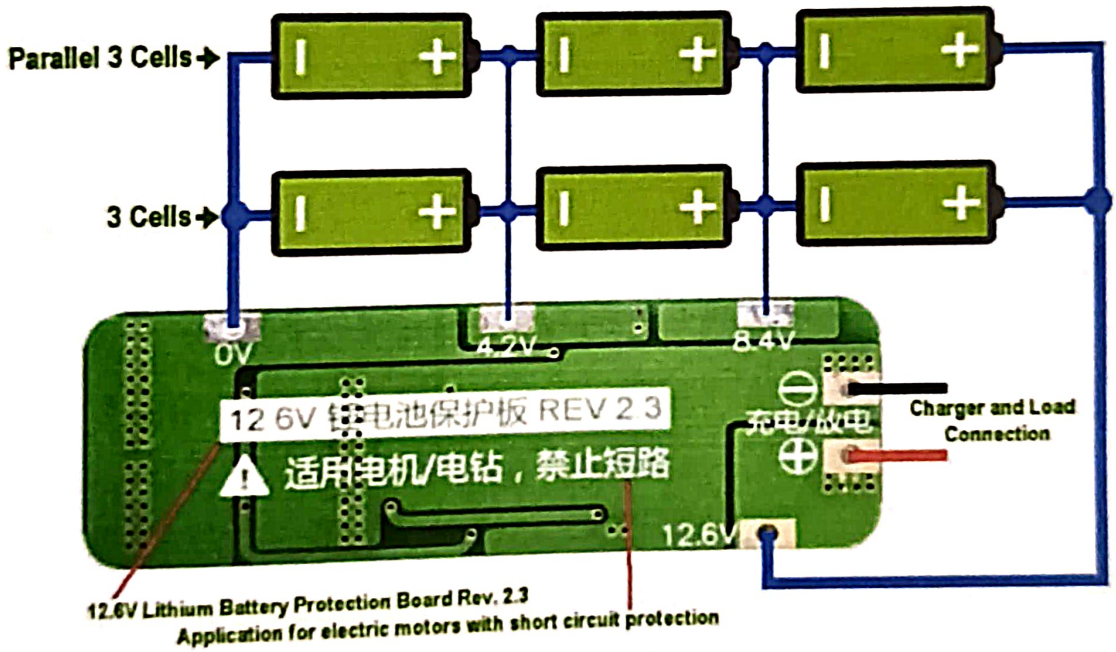

Fig: 6.8 BMS

A Battery Management System (BMS) is a crucial component in managing the performance, safety, and longevity of rechargeable batteries. The specifications of a BMS can vary depending on the application and the type of batteries it is designed to manage.A Battery Management System (BMS) is a crucial component designed to monitor, control, and safeguard rechargeable batteries. Its primary function is to ensure the efficient and safe operation of the battery pack. The BMS accomplishes this through a set of specifications and features.At its

core, the BMS oversees the individual cells within the battery pack, managing factors such as voltage, current, and temperature. The BMS also facilitates communication between the battery and the device it powers, ensuring optimal energy transfer and usage.

6.9 PRESSURE SENSOR

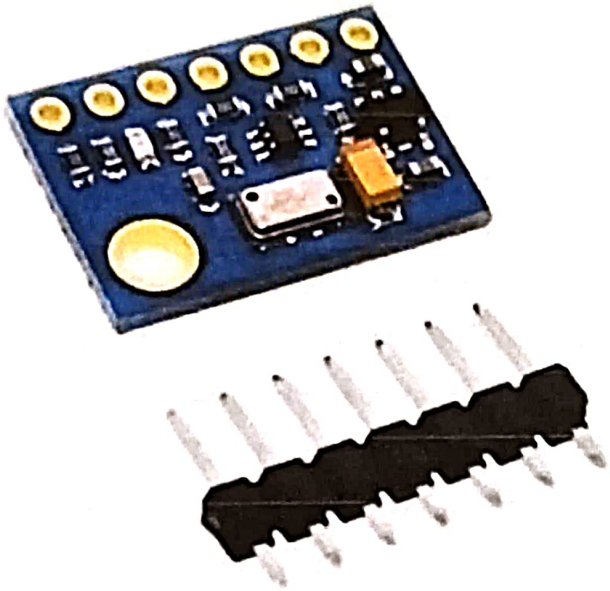

Fig: 6.9 Barometric pressure sensors

Barometric pressure sensors, also known as barometers, are electronic devices designed to measure atmospheric pressure. The specifications for a barometric pressure sensor may vary depending on the specific model and manufacturer, but here are some common specifications you might find:

- Pressure Range: Barometric pressure sensors are designed to measure atmospheric pressure within a certain range. Common ranges include 300 hPa to 1100 hPa.
- Accuracy: Accuracy is a crucial specification, indicating how closely the sensor's readings match the actual atmospheric pressure. It is usually expressed as a percentage of the full-scale range or in hPa.
- Resolution: Resolution refers to the smallest change in pressure that the sensor can detect. It is usually specified in terms of the smallest pressure difference that can be measured, such as in millibars (mbar) or pascals (Pa).

6.10 CAMERA SENSOR

In an era defined by technological innovation and rapid advancements, the realm of imaging continues to undergo transformative evolution. At the forefront of this revolution stands the 25-megapixel USB camera, a pinnacle of engineering excellence and a testament to the boundless possibilities of modern imaging technology. With a staggering resolution of 25 megapixels, this USB camera redefines the very essence of clarity and precision in visual capture. Each image is imbued with an unprecedented level of detail and definition, allowing for immersive exploration and meticulous analysis across a myriad of applications. Central to the versatility of the 25-megapixel USB camera is its seamless integration with a USB interface, ensuring effortless connectivity and compatibility with a diverse array of devices and systems. Whether utilized in scientific research, industrial inspection, or creative endeavors, this camera offers unparalleled convenience and flexibility in data acquisition and analysis.

One of the most remarkable features of this cutting-edge camera is its remarkable focus range, spanning from 4 to 20 meters. This expansive range empowers users to capture images with unparalleled clarity and depth, whether close or at a distance. From macro photography to long-range surveillance, the 25-megapixel USB camera delivers exceptional performance across a spectrum of scenarios.

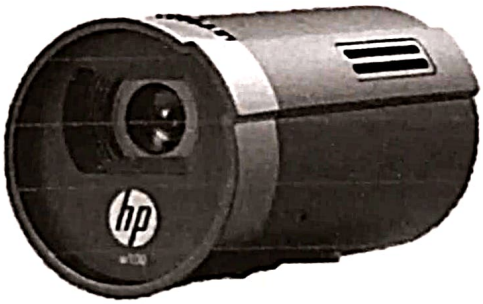

Fig 6.10 Camera

Beyond its remarkable technical specifications, the 25-megapixel USB camera embodies a commitment to innovation and excellence in imaging technology. Designed and engineered to exceed the expectations of even the most discerning users, this camera represents a convergence of cutting-edge hardware and sophisticated software, resulting in a seamless and immersive imaging experience.In conclusion, the introduction of the 25-megapixel USB camera heralds a new era of imaging excellence, where clarity, precision, and versatility converge to redefine the boundaries of visual exploration and analysis. Whether unlocking new insights in scientific research, enhancing productivity in industrial applications, or unleashing

creativity in artistic pursuits, this camera stands as a beacon of innovation, empowering users to capture the world with unprecedented clarity and depth. Some specifications of the selected camera are:

- Image resolution: 25 MP
- USB Interface
- Focus Range: 4 cm to 20m
- Image sensor quality CMOS (Complementary Metal-Oxide-Semiconductor) sensor

6.11 LIGHT MODULE

With a luminous flux of 180 to 200 lumens in radiant white, it offers unparalleled brightness while consuming just 53 watts of power, ensuring both exceptional performance and significant energy savings. Built to excel in various environments, it operates reliably in temperatures ranging from 0 to 60 degrees Celsius. Crafted with precision engineering, it withstands demanding conditions while delivering consistent illumination.

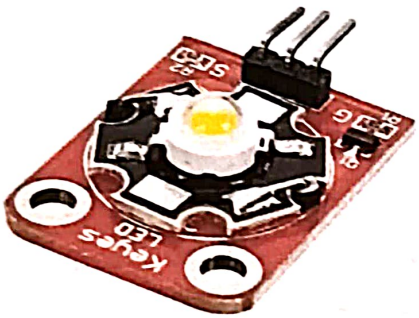

Fig 6.12

From indoor spaces to outdoor landscapes, our light module sets a new standard for lighting excellence. Its efficiency, reliability, and performance redefine illumination, offering a glimpse into a brighter, more sustainable future. Join us in embracing the next generation of lighting technology with our High-Efficiency Light Module.

Specifications

- Luminous flux: White 180-200LM (lumens)
- Voltage: 5V
- Power: 3W
- Viewingangle: 180°
- Working temperature: -20° to 60°

6.12 HULL

For the hull, we have decided to make it using a filter tube. Using a water filter as a hill for an underwater drone involves repurposing the filter's structure and functionality to create a stable platform for the drone to rest on or maneuver around underwater. The process begins with selecting a suitable water filter with a robust structure and adaptable design. The chosen filter should offer a stable foundation for the underwater drone and allow for easy modifications to accommodate additional components. Once a suitable water filter is chosen, it needs to be modified to accommodate the underwater drone. This may involve removing certain parts or sections of the filter to create a flat surface or platform where the drone can rest or move around. A secure mounting mechanism is essential to attach the drone to the modified water filter. This may involve using clamps, brackets, or other fastening devices to ensure the drone remains stable and secure while submerged.

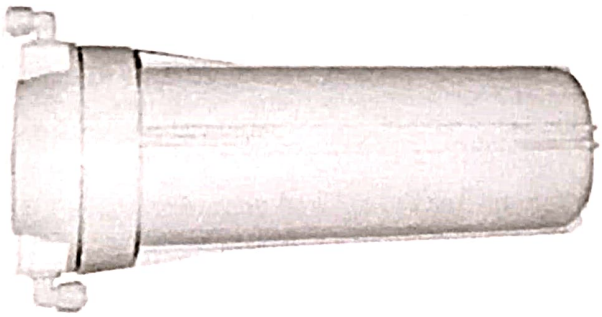

Fig 6.12 Hull

Adjusting the stability and buoyancy of the filter-drone assembly is crucial for successful underwater operation. Additional weights or flotation devices may be required to maintain balance and ensure the assembly remains level underwater. The control systems of the underwater drone, including propulsion, navigation, and communication, must be seamlessly integrated with the water filter hill. This involves routing cables or wiring through the filter's structure and ensuring all components are properly connected and functional.

Before deployment, thorough testing and calibration of the modified water filter and underwater drone assembly are essential. This ensures optimal performance, reliability, and safety underwater. Once testing is complete, the water filter hill and underwater drone assembly are ready for deployment in real-world underwater environments. Operators monitor the drone's performance closely and make any necessary adjustments to ensure safe and efficient operation during exploration, surveillance, or other underwater tasks.

Features:-Tensile Strength of 47 - 79 Mpa and a Melting Point of 130°C.

6.13 EPOXY RESIN

Epoxy resins are commonly used in constructing and encapsulating components in Remote Operated Vehicles (ROVs) due to their excellent adhesive properties, high strength, and resistance to water and chemicals. Here are some specific applications of epoxy resins in ROVs.

Encapsulation of Electronics: ROVs often have sensitive electronic components that need to be protected from water, pressure, and other harsh environmental conditions. Epoxy resins are used to encapsulate and protect these electronics, providing a water-tight seal and preventing corrosion. Cable and Connector Sealing: Epoxy resins are used to seal and protect cable connections and connectors on ROVs. This helps prevent water ingress, which is crucial for maintaining the integrity of electrical connections and ensuring the proper functioning of the vehicle. Pressure Hulls and Structural Components: Some ROVs have structural components, such as pressure hulls, made from composite materials that include epoxy resins. Epoxy provides high strength and durability, making it suitable for constructing components that need to withstand underwater pressure.

Buoyancy Materials: Epoxy resins can be used in the formulation of buoyancy materials for ROVs. These materials are designed to provide buoyancy while being resistant to water absorption. Epoxy-based buoyancy materials contribute to the overall stability and floatation of the ROV. Sensor Mounting and Integration: Epoxy resins are employed to mount and integrate various sensors onto the ROV structure. This includes sonar sensors, cameras, and other instrumentation. Epoxy provides a secure and durable bond while maintaining the desired sensor orientation.

CHAPTER 7

RESULT

The project yielded several key findings and outcomes. Firstly, the implementation of the Raspberry Pi-controlled system was successful, the integration of multiple BLDC motors with camera. The Python programs developed facilitated seamless communication and control, enhancing the overall user experience.

Our project has achieved

- Real-time monitoring underwater using a camera.
- Having a speed of 15cm/second and depth range of 10 meters.
- Creating a low cost underwater drone.

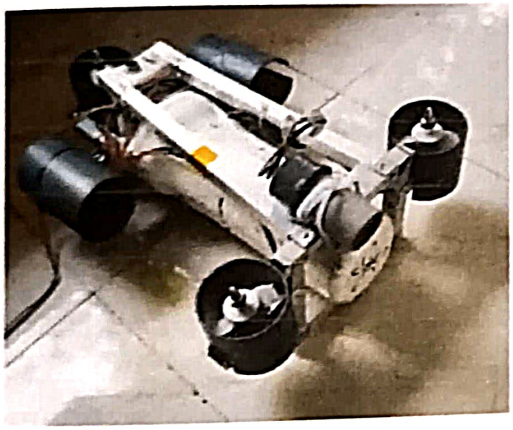

Fig 7.1 Underwater drone

Fig 7.2 Captured image

The Fig 7.1 shows the completed project and Fig 7.2 is the image captured by the underwater drone. We have identified a slight turbulence due to under currents in a flowing river, the battery is able to keep the drone functioning for 40 minutes after full charge. The visibility range of the drone is 1.5 meters in a dark environment or at night.

CHAPTER 8

CONCLUSION

The design and fabrication of the underwater drone (ROV) have yielded several key achievements and marked successes in meeting the outlined project objectives.

Key Achievements:

1. Functional Integration: Successful integration of components such as Pixhawk, Raspberry Pi, A2212 motor, and the 3S 2P battery configuration, leading to a cohesive and efficient ROV system.

2. Robust Mechanical Design: Implementation of a durable hull using epoxy resin, providing excellent waterproofing and structural strength, ensuring the protection of internal electronics in underwater environments.

3. Optimized Electrical System: Efficient power distribution and utilization of a 3S 2P battery configuration, managed by a Battery Management System (BMS), contributing to extended operational durations and enhanced safety.

4. Software Synergy: Successful deployment of Ardusub software for underwater vehicle control, showcasing the adaptability and capabilities of open-source platforms for ROV applications.

5. Comprehensive Testing: Thorough testing in various environments, validating buoyancy control, depth maintenance, and maneuverability, ensuring the ROV's reliability in diverse underwater scenarios.

Overall Success:

The ROV project has successfully met its objectives, providing a cost-effective and functional solution for underwater exploration and research. The integration of components, coupled with a robust hull design and efficient electrical and software systems, culminates in an underwater drone poised for various applications.

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APPENDIX

Program for Raspberry pi :

```
import serial
import time
import picamera
from io import BytesIO
# Configure the serial port
ser = serial.Serial('/dev/ttyACM0', 9600, timeout=1)
def forward():
   ser.write(b'F')
def back():
   ser.write(b'B')
def left():
   ser.write(b'L')
def right():
   ser.write(b'R')
def roll():
   ser.write(b'A')
def pitch():
   ser.write(b'Z')
def down():
   ser.write(b'D')
def up():
   ser.write(b'U')
```

```
def stop():
    ser.write(b'S')
def light_on():
    ser.write(b'W')
def light_off():
    ser.write(b'E')
def capture_image():
    with picamera.PiCamera() as camera:
        camera.resolution = (640, 480)
        camera.start_preview()
        time.sleep(2)  # Allow camera to warm up
        stream = BytesIO()
        camera.capture(stream, format='jpeg')
        stream.seek(0)
        return stream.getvalue()
# Example usage
try:
  while True:
```

Program for Arduino UNO:

```
#include <Wire.h>
#include <PulsePosition.h>
#include <Servo.h>
float RatePitch, RateRoll, RateYaw;
float RateCalibrationPitch, RateCalibrationRoll, RateCalibrationYaw;
int RateCalibrationNumber;
```

```
PulsePositionInput ReceiverInput(RISING);
float ReceiverValue[] = {0, 0, 0, 0, 0, 0, 0, 0};
int ChannelNumber = 0;
float Voltage, Current, BatteryRemaining, BatteryAtStart;
float CurrentConsumed = 0;
float BatteryDefault = 1300;
uint32_t LoopTimer;
float DesiredRateRoll, DesiredRatePitch, DesiredRateYaw;
float ErrorRateRoll, ErrorRatePitch, ErrorRateYaw;
float InputRoll, InputThrottle, InputPitch, InputYaw;
float PrevErrorRateRoll, PrevErrorRatePitch, PrevErrorRateYaw;
float PrevItermRateRoll, PrevItermRatePitch, PrevItermRateYaw;
float PIDReturn[] = {0, 0, 0};
float PRateRoll = 0.6;
float IRateRoll = 3.5;
float DRateRoll = 0.03;
float PRatePitch = PRateRoll;
float IRatePitch = IRateRoll;
float DRatePitch = DRateRoll;
float PRateYaw = 2;
float IRateYaw = 12;
float DRateYaw = 0;
float MotorInput1, MotorInput2, MotorInput3, MotorInput4;
char command;
Servo esc1;
```

```
Servo esc2;
Servo esc3;
Servo esc4;
Servo esc5;
void battery_voltage() {
 Voltage = (float)analogRead(15) / 62;
 Current = (float)analogRead(21) * 0.089;
}
void read_receiver() {
 ChannelNumber = ReceiverInput.available();
 if (ChannelNumber > 0) {
  for (int i = 1; i <= ChannelNumber; i++) {
   ReceiverValue[i - 1] = ReceiverInput.read(i);
  }
 }
}
void gyro_signals() {
 // Add your existing gyro signal code here
}
void pid_equation(float Error, float P, float I, float D, float PrevError, float PrevIterm) {
 // Add your existing PID equation code here
}
void reset_pid() {
 // Add your existing reset PID code here
}
```

```
void setup() {
 pinMode(5, OUTPUT);
 digitalWrite(5, HIGH);
 pinMode(13, OUTPUT);
 digitalWrite(13, HIGH);
 Wire.setClock(400000);
 Wire.begin();
 delay(250);
 // Add your existing setup code here
 // Servo initialization
 esc1.attach(9);
 esc2.attach(3);
 esc3.attach(10);
 esc4.attach(5);
 esc5.attach(6);
 esc3.writeMicroseconds(1500);
 esc5.writeMicroseconds(1500);
 esc2.writeMicroseconds(1500);
 esc4.writeMicroseconds(1500);
 esc1.writeMicroseconds(1500);
 delay(3000);
 // Receiver input initialization
 ReceiverInput.begin(14);
 while (ReceiverValue[2] < 1020 || ReceiverValue[2] > 1050) {
  read_receiver();
```

```
    delay(4);
   }
   LoopTimer = micros();
  }
  void loop() {
   // Add your existing loop code here
   // Servo control based on serial input
   if (Serial.available() > 0) {
    command = Serial.read();
    STOP();
    switch (command) {
     case 'F':
      forward();
      break;
     case 'B':
      back();
      break;
     case 'L':
      left();
      break;
     case 'R':
      right();
      break;
     case 'A':
      ROLL();
```

```
        break;
      case 'Z':
        PITCH();
        break;
      case 'D':
        DOWN();
        break;
      case 'U':
        UP();
        break;
      case 'S':
        STOP();
        break;
      case 'W':
        LIGHT_ON();
        break;
      case 'E':
        LIGHT_OFF();
        break;
    }
  }
}
void forward() {
  esc5.writeMicroseconds(1400);
  esc4.writeMicroseconds(1400);
```

```
}
void back() {
 esc5.writeMicroseconds(1600);
 esc4.writeMicroseconds(1600);
}
void left() {
 esc5.writeMicroseconds(1400);
 esc4.writeMicroseconds(1600);
}
void right() {
 esc5.writeMicroseconds(1600);
 esc4.writeMicroseconds(1400);
}
void PITCH() {
 esc1.writeMicroseconds(1400);
 esc2.writeMicroseconds(1600);
 esc3.writeMicroseconds(1400);
}
void ROLL() {
 esc1.writeMicroseconds(1600);
 esc2.writeMicroseconds(1400);
 esc3.writeMicroseconds(1600);
}
void DOWN() {
 esc1.writeMicroseconds(1600);
```

```
  esc2.writeMicroseconds(1400);
  esc3.writeMicroseconds(1400);
 }
 void UP() {
  esc1.writeMicroseconds(1400);
  esc2.writeMicroseconds(1600);
  esc3.writeMicroseconds(1600);
 }
 void STOP() {
  esc1.writeMicroseconds(1500);
  esc2.writeMicroseconds(1500);
  esc3.writeMicroseconds(1500);
  esc4.writeMicroseconds(1500);
  esc5.writeMicroseconds(1500);
 }
 void LIGHT_ON() {
  digitalWrite(13, HIGH);
 }
 void LIGHT_OFF() {
  digitalWrite(13, LOW);
 }
```